

FC → SC Plan — Meeting Notes

Direction

Swap existing SC → FC pipeline to run **FC** → **SC**. Build on the current evaluation framework. Use demeaned correlation as headline metric.

Metric suite ([From Slideshow](#)) (Also from Zalesky's paper)

- Raw correlation
- Demeaned correlation (headline — subject-specificity)
- Average rank
- Top-1 accuracy
- Compared against Null (PCA-only) and Oracle (autoencoder) bounds
- Reported over 10 train/val/test splits

Phase 0 — Swap direction + baseline

- Flip PCA + PLS + projector code: FC in, SC out
- Run all baselines (PCA+PLS variants, Krakencoder, Sarwar MLP, Chen GNN) reversed
- HCP n=957, family-aware splits

0.5 — SC specificity Metric (Written as 0.1 on whiteboard, I think this is correct)

- Compute demeaned correlation
- Healthy (~0.10 like SC→FC) → proceed to Phase 1
- Near-zero → pivot, bet on Phase 1 recovering signal

Phase 1 — Multi-task FC concatenation

- Inputs: rest + 7 HCP tasks (working memory, gambling, motor, language, social, relational, emotion)
- Vectorize each (~60k edges), PCA per condition, concatenate, linear layer, $\text{PCA}^{-1} \rightarrow \text{SC}$
- Variants: per-condition vs joint PCA; rest-only vs +1 task vs +all; equal vs learned task weighting; linear fusion vs GNN
- Headline: trajectory of demeaned correlation as tasks added
- Secondary: per-region demeaned correlation → task-to-region specificity map

Phase 2 — Time-series transformer

- Input: $R \times T$ tensor per task (regions $\times$ time)
- Per-task weights or task embeddings (W_{rest} , W_{task1} , ...)
- Aggregation across tasks $\rightarrow$ PCA-SC space $\rightarrow$ $\text{PCA}^{-1} \rightarrow$ SC
- Formal: $\Pr(X^{\text{SC}}_{\text{test}} | \sum X^{\text{FC}}_{\text{test},i}, X^{\text{SC}}_{\text{train}})$
- Variants: shared transformer + embeddings vs separate per task; window length; pooling method
- Headline: cumulative demeaned correlation trajectory across Phase 0 $\rightarrow$ 1 $\rightarrow$ 2

Phase 2.5 — optional 3T vs 7T

- Run transformer on HCP 7T (n=184)
- Saturation test: model limit vs data limit

Phase 3 — Unified bidirectional model (longer-term)

- JEPA-style backbone
- Any-modality-to-any-modality, self-supervised
- Out of scope for current timeline; mention as future direction

Data ??

This part I am the most lost on I don't remember the exact datasets you used here is what I remember

- HCP Young Adult 3T, n=957
- HCP Retest n=46 (reliability ceiling)
- HCP n=184 7T and 3T (Phase 2.5 only) from UK BioBank - optional

Possible Outcomes (From my imagination)

- **A** — monotonic demeaned correlation increase across phases $\rightarrow$ hypothesis supported
- **B** — plateau after Phase 1 $\rightarrow$ static FC sufficient, transformer unnecessary
- **C** — flat from Phase 0 $\rightarrow$ FC lacks subject-specific SC signal, data quality bottleneck
- **D** — non-monotonicity / interactions in Phase 2 $\rightarrow$ richer finding, follow-up
- **E** — task-to-region specificity in per-region analysis $\rightarrow$ biological headline regardless of phase

Open questions

- Per-task vs joint PCA?

- Numerical threshold for 0.1 gate, or qualitative?
 - Phase 2 transformer: shared + embeddings, or separate per task?
 - 7T data preprocessed and ready?
 - Target venue?
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